

IFS-Light: An Interactive Framework for Single-view Face Relighting with both Facial and Lighting Consistency

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Abstract

Single-view face relighting aims to adjust the portrait lighting while preserving the original background. Although recent diffusion-based methods achieve great relit results by using reference lighting and facial features as conditions for the diffusion relighting process, they are limited by the incompleteness of these conditions, such as the absence of explicit constraints on skin tone and the lack of spatial coverage for hard shadows, which results in facial and lighting inconsistencies. To address these challenges, we propose IFS-Light, an interactive framework that leverages spatial-nonspatial conditioning mechanism to localize facial features and reference lighting, then optimize their interplay in the relighting process. To ensure facial consistency, we first combine skin-tone-scaled conditions with shape information for tone adjustment, enhanced by a detail mask that identifies modifiable facial regions. Skin and shape parameters are then optimized to preserve both skin tone and fine details. To maintain lighting consistency, we propose a ray-tracing-based formulation that decomposes reference lighting into diffuse and non-diffuse components. These, integrated with shape information, assist in positioning light and shadow regions. Both components are then encoded for precise control over color and intensity. In addition, we propose an innovative and user-friendly solution for adjusting light conditions, which enables the user to precisely adjust the position of the light source to flexibly control both light intensity and direction, thereby making it easier to achieve the desired relighting results. Extensive experiments show that IFS-Light achieves superior relighting results compared to state-of-the-art methods. The code and appendix are available here.

CCS Concepts

• Computing methodologies → Image processing.

Keywords

Single-view Face Relighting; Facial and Lighting Consistency

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1 Introduction

Single-view face relighting aims to adjust lighting in portraits while preserving the original background content, with applications in film production, portrait editing, image synthesis, and augmented reality [15, 25]. However, due to the deep coupling of lighting and facial features in the input and reference images, accurately modeling physical interactions of reference lighting and input facial features in the relit portrait presents significant challenges [25, 36].

Most conventional methods [1, 15, 17, 19, 20, 24, 31, 32, 44] approach these challenges by decomposing the input and reference images into albedo and shading, and then recomposing them to generate relit images, as shown in Fig. 1 (a). However, due to the ill-posed nature of intrinsic decomposition, the facial features from the albedo change, which leads to facial inconsistency. In addition, they focus on enhancing the comprehensiveness of reference lighting in the shading, typically employing spherical harmonic (SH) parameters [28] or hard shadow maps. SH parameters are primarily effective for capturing diffuse lighting effects, including ambient lighting and soft shadows [15, 24], while shadow maps are better suited for non-diffuse effects like dominant directional lighting and hard shadows [25]. However, the above representations fail to account for both lighting effects, such as subtle shadow changes in case of bold shadows commonly seen in portraits. This limitation further results in noticeable lighting inconsistencies, and when combined with altered facial features, produces unnatural color artifacts, such as the reddish skin tone in Fig. 2 (a).

To maintain facial consistency and lighting completeness, recent diffusion-based methods [2, 25, 37] have reformulated single-view face relighting as a conditional generation process. Unlike conventional methods, which rely heavily on the accuracy of intrinsic decomposition, these methods use reference lighting and input facial features as conditions for the diffusion process, optimizing them during relighting generation, as illustrated in Fig. 1 (b). However, the design of facial features and lighting in diffusion-based methods has limitations. For example, these methods use facial embeddings [25] or cross-color ratios (CCR) [11] of the surface [37] to maintain skin tone and facial details. Intuitively, they ignore skin tone constraints, which are crucial for facial consistency, potentially resulting in undesirable color shifts, such as greyish tones.

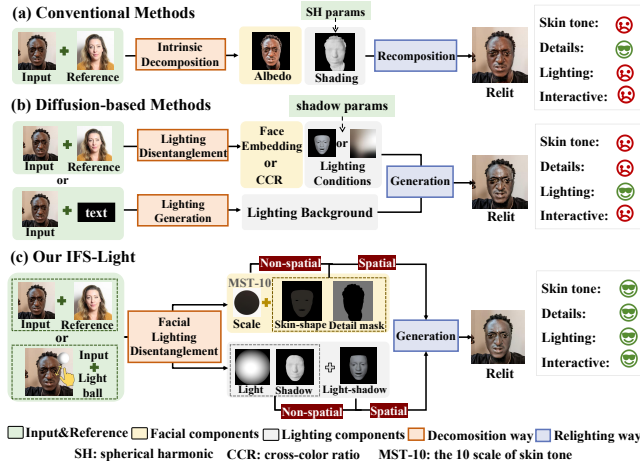


Figure 1: Pipeline comparison of single-view face relighting: (a) conventional methods, (b) diffusion-based methods, and (c) our IFS-Light, an interactive framework through spatial-nonspatial conditions with both facial and lighting consistency. Note that input image comes from MST-E dataset and the reference image comes from Flickr [8].

Additionally, although background information is used as a spatial condition, it can interfere with the relighting process and cause distorted details. For lighting conditions, these methods often rely on shadow intensity parameters and shading references from SH [25], or illumination maps derived from light sources [37]. However, they lack spatial control over hard shadows, making it difficult to accurately define their coverage, leading to arbitrary shadow artifacts, as shown in the zoomed-in areas of Fig. 2 (b).

To address the above issues, we propose a single-view face relighting framework named IFS-Light, which ensures facial consistency through skin-shape guided tone adjustment and lighting consistency via ray-tracing-based light-shadow guided spatial coverage. The core idea of the above designs is the spatial-nonspatial condition mechanism: spatial conditions enable the accurate positioning of facial features and reference lighting, whereas non-spatial conditions guide the optimization their precise interplay. Both sets of conditions are harmoniously integrated into the diffusion process to facilitate the generation of relit images. Specifically, to ensure facial consistency, we introduce skin-shape conditions, where skin tone is predicted via a tone-scaling model. We incorporate spatial conditions for localized tone adjustments and a detail mask for refines editable facial regions while maintaining background integrity. The skin and shape parameters are then encoded to preserve natural skin tone and details. To provide complete reference lighting, we design a light-shadow condition predicated on a ray-tracing-based formulation that decomposes reference lighting into diffuse and non-diffuse terms. By integrating these with shape information, we first generate spatial conditions to localize light and shadow regions. Both terms are then encoded as non-spatial conditions for precise control over color and intensity.

Furthermore, to enhance the application of single-view relighting, improving interactivity is essential. The common approach

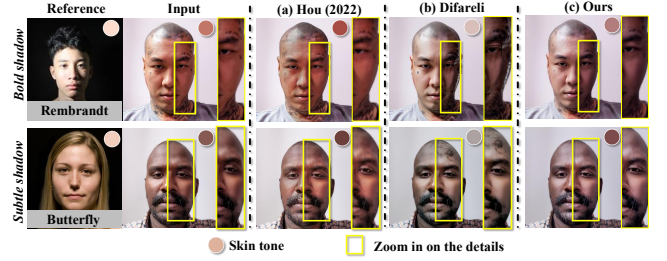


Figure 2: Prevailing methods exhibit facial and lighting inconsistencies in visualizations when compared to IFS-Light. It should be noted that input images are sourced from MST-E dataset and the reference images are from Flickr [8].

involves user-provided reference images, which needs exhaustive searches for a specific style [23]. To reduce this limitation, some methods offer optional parameters, such as SH [17, 31, 44], shadow intensity [25], or the use of text descriptions [2] to replace images, allowing users to adjust the reference lighting, as shown in Fig. 1. However, they do not enable free adjustment of the light source. To address this, we propose a solution that enables users to adjust the position of the light source (light ball), providing flexible control over light intensity and direction, thereby enhancing interactive relighting. Our key contributions are summarized as follows:

- We introduce IFS-Light, an interactive framework that employs spatial-nonspatial condition mechanism for consistent relighting. Furthermore, we provide flexible user control, allowing users to easily design portrait lighting effects.
- We ensure facial consistency through skin-shape guided tone adjustment, enhanced by a detail mask to identify regions for modification, and maintain lighting consistency via ray-tracing-based light-shadow guided spatial coverage.
- Compared to prevailing methods, IFS-Light presents state-of-the-art (SOTA) relighting performance with two additional consistency metrics on datasets. Moreover, our interactive way provides a new roadmap for community.

2 Related Works

2.1 Conventional Methods

2.1.1 Lighting Representation. Existing face relighting methods typically adopt two lighting representations [16]: SH parameters [1, 15, 17, 19, 20, 31, 32, 44] or hard shadow maps [15, 24]. SH is the most widely used representation [1, 15, 19, 20, 31, 32]. Earlier methods decompose albedo and shading based on Lambertian surfaces and SH parameters via CNN [19, 20, 31, 32] or regularized optimization [1], which *suffer from* lighting inconsistencies and cannot handle non-diffuse effects like hard shadows. Zhou et al. [44] bypass explicit intrinsic decomposition by directly estimating target luminance from source luminance and target lighting. *However*, it is limited to soft shadows due to ambient lighting training.

Recent methods address non-diffuse lighting by modeling hard shadow maps. Nestmeyer et al. [24] use a binary visibility map from diffuse rendering to model hard shadows, the indirect constraints by face geometry can *result in* shadows with inconsistent shapes.

Hou et al. (21) [17] compute shadow maps on a morphable model using ray tracing, emphasizing hard shadow borders, yet *lack* of geometric information can result in arbitrary shadow shapes. In Hou et al. (22) [15], they predict shadow masks via ray tracing and generate relit images using CNN-derived albedo and shading. *However*, their focus on non-diffuse effects overlooks soft shading, limiting performance under ambient lighting.

Additionally, while existing methods focus on resolving lighting inconsistencies, they often *fail* to maintain facial consistency. This stems from the ill-posed nature of intrinsic decomposition and is further exacerbated by challenges in varying scene lighting conditions [23, 25]. **Different from them**, our IFS-Light avoid intrinsic decomposition and integrates skin conditions into the diffusion process. Additionally, our lighting representation combines ambient lighting with both soft and hard shadows, enabling handling of diffuse and non-diffuse lighting effects, as in Tab. 1.

2.1.2 Shadow Editing. Another closely related area to IFS-Light is shadow editing, which focuses on facial shadow manipulation, including shadow removal [14, 22, 38, 40, 43], softening [9], and modification [16]. While shadow removal techniques [22, 40, 43] aim to eliminate shadows entirely, they *lack* the ability to control the degree of shadow softening. In contrast, shadow softening methods [9] provide adjustable control over shadow intensity. Similarly, COMPOSE [16] supports advanced shadow manipulations, such as intensifying, reshaping, and rotating shadows. However, these methods are *limited* to edit shadowed regions and do not adjust broader lighting conditions. **Unlike these methods**, IFS-Light jointly models lighting and shadows using a ray-tracing-based formulation, enabling more flexible relighting.

2.2 Diffusion-based Methods

With the advancement of diffusion models, diffusion-based generative methods have enabled creative results in face relighting. Methods like Relightful Harmonization [29], IC-Light [41] and LuminaBrush [33] leverage background to define lighting conditions, synthesizing portraits harmonized with the background. Conversely, SynthLight [4] and DiffRelight [13] focus on portrait-based relighting, employing a diffusion model that learns to re-render relighting faces. *However*, these methods often modify the original scene, limiting their applicability to single-view face relighting.

Other methods [2, 25, 37] preserve the original scene while avoiding intrinsic decomposition by employing facial and lighting conditions to steer the diffusion process for relighting. Difareli [25] disentangles SH and shadow intensity from the target lighting, using a shading reference rendered from SH as lighting conditions and incorporating face embeddings to preserve facial features. Retinex-Diffusion [37] applies a multi-scale Retinex model [27] with 2D Gaussian kernels to extract illumination maps and computes the CCR [11] of the surface to maintain geometry. Text2Relight [2] uses text descriptions for lighting conditions, separating foreground and background during training to maintain subject consistency.

However, these methods all *suffer from* facial and lighting inconsistencies. Regarding lighting inconsistency, Difareli [25] only encodes soft shadows in its reference, limiting its ability to model the spatial coverage of hard shadows. Retinex-Diffusion [37] adjusts global color and intensity but lacks fine-grained control over facial

Table 1: Comparison of single-view face relighting methods.

Methods	Facial Consistency		Lighting Consistency	Interactivity
	Skin Tone	Details		
Conventional	None	Albedo	Diffuse	Parameters
Diffusion-Based	None	Face embeddings	Diffuse	Parameters
Ours	Skin tone scale	Detail mask	Diffuse and non-diffuse	Light Ball

lighting. Text2Relight [2] often faces uncontrolled color biases and distortions due to subjective inputs. Additionally, the lack of explicit skin tone constraints can cause artifacts, especially with complex facial features. **In contrast**, our IFS-Light framework integrates non-spatial and spatial skin conditions with a detail mask to ensure facial consistency. It offers precise control over hard shadow spatial coverage through light-shadow guidance, as in Tab. 1.

2.3 Interactive Way

The most common interactive approach for single-view face relighting methods [2, 17, 25, 31, 44] relies on reference images provided by users, but this often lacks precise control over the lighting and needs extensive searches to match a specific style [23]. To alleviate the limitation, some methods offer additional parameters, such as SH [17, 31, 44] and shadow intensity [25], to adjust the reference lighting. *However*, these methods do not allow free adjustment of the light source, thus limiting the interactivity. Text2relight [2] replaces reference images with text descriptions, *but* the subjective nature of text generation may not accurately generating the desired lighting conditions. **Unlike them**, we introduce a solution that enables users to adjust the position of the light source, providing flexible control over the intensity and direction of lighting, thereby facilitating more interactive and accurate relighting, as in Tab. 1.

3 Method

3.1 Problem Formulation

Given an input portrait P_o with lighting condition L_o , the single-view face relighting methods aim to build a model $\mathcal{T}_\theta$ that can replace L_o with L_t from a reference portrait P_t :

$$P_o^{L_t} = \mathcal{T}_\theta(P_o^{L_o}, P_t^{L_t}). \quad (1)$$

Conventional methods [1, 15, 17, 19, 20, 24, 31, 32, 44] typically formulate Eq. 1 as a de-recomposition pipeline. However, as discussed in Sec.1 and Sec.2, the ill-posed nature of intrinsic decomposition and the incomplete representation of reference lighting cause facial and lighting inconsistencies (Fig. 2 (a)). Diffusion-based methods [2, 25, 37] address these limitations by using reference lighting and input facial features as conditions in the diffusion process, optimizing them during the relighting generation. However, the lack of skin constraints and explicit spatial control over hard shadows still leads to these inconsistencies (Fig. 2 (b)).

To address these, we propose IFS-Light, which incorporates non-spatial and spatial condition mechanisms specifically designed for facial attributes and lighting conditions, as shown in Fig. 3 (a). We reformulate Eq. 1 in a three-step process:

$$SSG_o, M_o, f_o = \mathcal{G}_1(P_o^{L_o}), \quad (2)$$

$$LSG_t, l_t = \mathcal{G}_2(P_t^{L_t}), \quad (3)$$

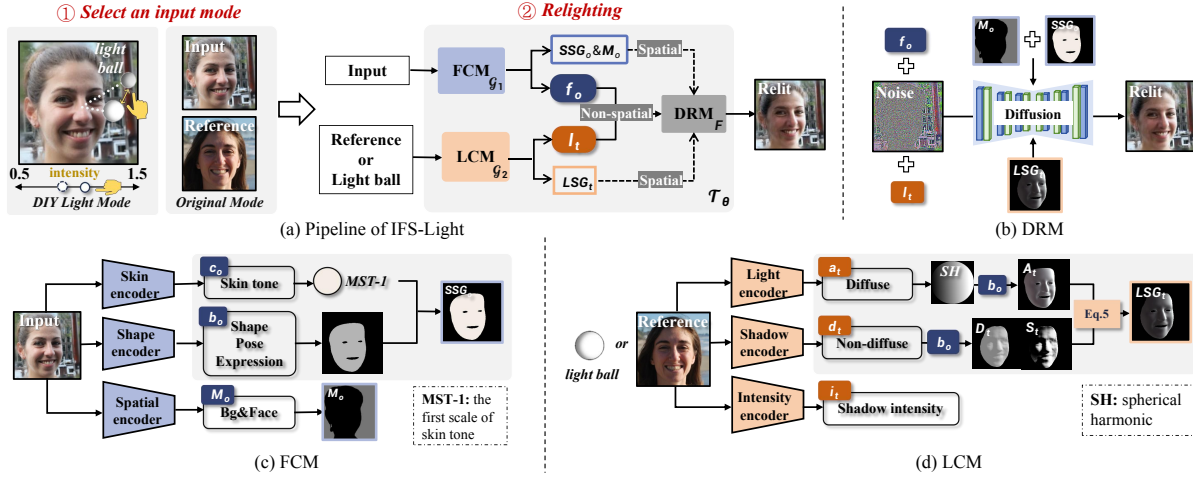


Figure 3: Overview of IFS-Light framework. First, users select an input mode: the interactive mode or the original mode. Next, the selected input is passed through the relighting pipeline, which consists of three core components: the Facial Conditions Module (FCM), the Lighting Conditions Module (LCM), and the Diffusion-based Relighting Module (DRM).

$$P_o^{Lt} = \mathcal{F}(SSG_o, M_o, LSG_t, f_o, l_t), \quad (4)$$

where $\mathcal{G}_1(\cdot)$ is the Facial Conditions Module (FCM), which decomposes non-spatial facial conditions, including skin color, into f_o and spatial conditions, including skin-shape guidance SSG_o and the detail mask M_o . $\mathcal{G}_2(\cdot)$ is the Lighting Conditions Module (LCM), which extracts non-spatial lighting conditions l_t and spatial conditions, light-shadow guidance LSG_t . Finally, $\mathcal{F}(\cdot)$ is the Diffusion-based Relighting Module (DRM), which implements the generative relighting function, as shown in Fig. 3 (b). In addition, to enhance interactivity for better application, our framework can support two input modes: an interactive mode to control the light ball and the original mode using the reference images provided by users.

3.2 Model Implementation

3.2.1 Facial Conditions Module. As shown in Fig. 3 (c), to implement the function $\mathcal{G}_1(\cdot)$ in Eq. 2, we propose the Facial Conditions Module (FCM), which aims to generate cohesive facial conditions from an input portrait. The facial conditions consist of spatial conditions — skin-shape guidance SSG_o and the detail mask M_o — along with non-spatial conditions, namely the facial features $f_o = \{c_o, b_o\}$. The spatial conditions guide $\mathcal{F}(\cdot)$ in locating the position of face, while c_o captures the skin tone and b_o represents the basic facial features, helping to further enhance fine details.

Non-spatial Skin Conditions. Skin tone representation is crucial for preserving facial consistency. Existing diffusion-based methods [25, 37] utilize face embeddings or compute the CCR of the surface to mitigate skin tone bias. However, such solutions lack precise control over skin tone adjustments and may lead to undesired variations in skin tone. To address these limitations, instead of directly decomposing the skin tone from albedo, we utilize skin tone scale c_o as a non-spatial condition to maintain consistency in the relighting process. Specifically, we leverage the Monk Skin Tone (MST) scale [30] as a benchmark and develop a skin tone prediction

model using the MST-E dataset [30] to classify the skin tones of input portraits into 10 distinct scales. Considering MST-E dataset consists of 3175 available training and testing images obtained from portraits and videos, we implement the skin tone scaling model by a lightweight model named RepViT [35] to estimate the skin tone scale of inputs, which then is encoded as non-spatial skin conditions. This approach ensures a consistent skin tone representation across different lighting conditions.

Spatial Skin-shape Guidance. To further provide geometric and textural guidance for skin tone and facial details, we generate the spatial skin-shape guidance SSG_o by integrating predefined skin tone scales c_o with base features b_o (Fig. 3 (c)). To capture the base feature b_o , such as 3D face shape, pose, and camera parameters for rendering, we employ DECA [7], which effectively predicts these parameters from a given image. Given b_o , we convert it into a triangle mesh using the three linear bases of the FLAME model [21] and retain only the face region. By setting c_o as the surface colors, we render this mesh according to the camera parameters to generate the skin-shape guidance. Guidance in diffusion models enhances interpretability by explicitly modeling the spatial information of skin tone and base features, enabling more accurate skin tone adjustments and reducing unwanted variations.

Spatial Detail Mask. While some diffusion-based methods, such as Difareli [25], have shown promising results by using background information as a conditional input, we find that this can hinder the model in preserving subtle facial details, *i.e.*, accessory and self-brought items, as shown in Fig. 4. Moreover, during the forward process of the diffusion model, the inherent noise carries background information, as illustrated in Fig. 3 (b), allowing the reconstruction of the background even without explicit input. To address these limitations, we propose employing a detail mask M_o excluding background information to delineate the boundary between the face and the background. As illustrated in Fig. 4, the grey

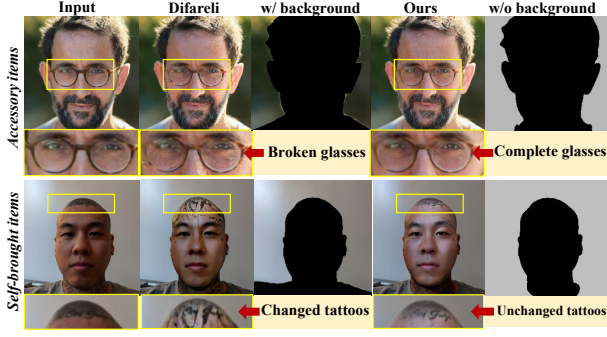


Figure 4: Comparison between Difareli (w/ background) and Ours (w/o background). Note that first input image is sourced from Unsplash [34] and the second is from MST-E dataset.

region indicates areas that should remain unchanged, while the black region marks the face area where changes are required. This detail mask enables the model to better understand and preserve high-frequency details, ensuring more accurate results.

In summary, our FCM combines a non-spatial skin tone scale with spatial skin-shape guidance and a detail mask that segments facial regions to remain unchanged. This integration enables more consistent and controllable relighting outcomes, effectively preserving skin tone across various scales while maintaining essential facial details, such as glasses and tattoos.

3.2.2 Lighting Conditions Module. As shown in Fig. 3 (d), to implement $\mathcal{G}_2(\cdot)$ in Eq. 3, we propose the Lighting Conditions Module (LCM), which aims to generate complete lighting conditions from the reference lighting based on a physical ray-tracing formulation to fully capture both diffuse and non-diffuse components. The conditions consist of spatial conditions, namely light-shadow guidance LLG_t , which helps locate areas of light and shadow, and non-spatial conditions, namely $l_t = \{a_t, d_t, i_t\}$. a_t captures the diffuse terms, such as ambient lighting with soft shadows, d_t captures the non-diffuse terms, such as dominant directional lighting with hard shadows, and i_t represents the shadow intensity. Together, these terms help optimize the realized lighting effects.

Non-spatial Lighting Conditions. Inspired by the work of Gao et al. [10], we propose a prior that separates the incident light into global and local components based on ray tracing. The sampled light from direction ω_i can be represented as follows:

$$L(\omega_i) = V(\omega_i) \cdot L_{global}(\omega_i) + L_{local}(\omega_i), \quad (5)$$

where the global light term $L_{global}(\omega_i)$ is the ambient lighting and produces soft shadows, the local light term $L_{local}(\omega_i)$ is the dominant directional lighting and produces hard shadows. $V(\omega_i)$ represents the visibility term based on ray tracing, also known as the shadow mask. A value of 1 indicates the light area, while 0 signifies a shadowed region.

Specifically, we utilize light and shadow encoders to predict a_t and d_t , representing the global and local lighting components, respectively. To effectively capture a_t , we employ DECA [7], which can predict SH parameters from a given image. For capturing the geometry-consistent non-diffuse terms d_t , we adopt Hou et al. [15],



Figure 5: Visual Comparison of the outputs with lighting guidance (right-bottom) from 300 to 1000 steps between Difareli and IFS-Light. Note that the input images come from MST-E dataset.

which uses ray tracing to extract hard shadow components. Additionally, to accurately estimate the shadow intensity i_t , inspired by Difareli [25], we fine-tune DiffAE [26] to predict shadow degree through latent codes, incorporating it as a non-spatial condition to further enhance the relighting process.

Spatial Light-shadow Guidance. To accurately capture the complex interplay of light and shadow in relighting, we propose generating light-shadow guidance LSG_t by combining lighting conditions l_t and base features b_o to convey spatial lighting information. As shown in Fig. 3 (d), we first render the 3D shape and apply a_t to it using the standard SH equation, generating an ambient lighting mask with soft shadows A_t . Next, we visualize d_t as a shadow mask S_t with a hard shadow intensity map D_t and apply Gaussian blur to smooth its edges, ensuring a more natural transition between shadowed and non-shadowed regions. Finally, based on the formulation in Eq. 5, we combine A_t , S_t , and D_t to generate light-shadow guidance. The rendering can be expressed as:

$$LSG_{i,j} = S_{i,j} \odot A_{i,j} + D_{i,j}, \quad (6)$$

where i, j denote the pixel (i, j) in image space. By embedding this guidance into the diffusion process, our IFS-Light gains light-aware relighting capabilities, enabling it to learn and adapt to variations in light and shadow dynamics rather than merely color shifts. This allows the production of relit portraits with realistic effects, including soft shadows, hard shadows and inter-reflections. For instance, in Fig. 5, Difareli [25] generates arbitrarily shaped results when modifying the light and shadow components. In contrast, our IFS-Light can model the physical interactions between geometry and lighting, guided by the light-shadow information, under varying skin tones and lighting conditions.

3.3 DIY Light

For better application in single-view face relighting, unlike prior methods that rely on optional parameters, we propose an interactive solution that allows users to freely adjust the light source position.

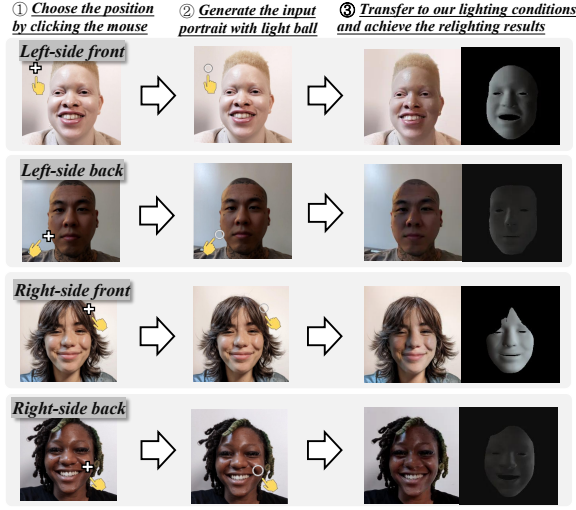


Figure 6: Interactive way and relighting results of using our IFS-Light. Note that input images are from MST-E.

As shown in Fig. 6, our interactive way follows three steps: first, users can freely select the light source position in the input image, adjusting both angles and distances by clicking the mouse. Then, our framework generates the input portrait with a light ball, which represents the light source. Finally, we transfer the lighting intensity and direction from the selected position to our lighting conditions, guiding the diffusion process and achieving the relighting results. This approach provides flexible control over lighting conditions, making it easier to achieve the desired lighting and shadow effects.

To effectively transfer to our lighting conditions, we first compute the lighting intensity based on the distance between the user’s set light ball and the center of the input image (DIY Light Mode in Fig. 3 (a)). We assume that when the lighting ball is placed at the center, the intensity is 1.5. As the distance increases, the intensity gradually decreases, reaching 0.5 at a certain distance. Next, we compute the angle between the light ball and the center point of the image. We have pre-collected basis vectors $\{a_t, d_t, i_t\}$, which represent our lighting model for eight predefined angles: N, S, E, W, NE, NW, SE, and SW. Specifically, we define N as the front-facing light, S as the back-facing light, E as the left-side light, and W as the right-side light. A linear operation, involving both the angle and intensity, is then performed to obtain the modified lighting vectors $\{a'_t, d'_t, i'_t\}$. Additionally, we provide settings for scaling factors of lighting and shadow intensity to enable more flexible relighting. After that, we generate spatial-nonspatial lighting conditions, which then directs the diffusion process to achieve the relighting results.

4 Experiment

4.1 Experimental Setups

Datasets. Following prior works [15, 25], we trained our conditional diffusion model on the FFHQ dataset [18], which includes 70k face images (60k for training and 10k for testing). Quantitative evaluations and ablation studies were conducted on the Multi-PIE

dataset [12], which contains 337 subjects captured under 19 different lighting conditions. To evaluate facial and lighting consistency, we conducted experiments on the Multi-PIE [12], FFHQ [18] and MST-E [30] datasets, selected for their diversity in gender, age, skin tone, facial details, and lighting conditions.

Evaluation Metrics. We evaluated our results by publicly used metrics: LPIPS [24] for perceptual quality, DSSIM [42] for structural dissimilarity, and MSE. Additionally, we employed two consistency metrics: CSIM, which uses cosine similarity between ArcFace [5] features to assess identity preservation, and Individual Typology Angle (ITA) to measure skin tone retention [3, 6]. All metrics are calculated by comparing each relit image to ground truth, focusing solely on the facial region through a face parsing method [39].

Implementation Details. Experiments were tested on a NVIDIA A6000 GPU. The training phase required approximately 4 days with a batch size of 16. The diffusion process used 1000 steps, with a learning rate of $1e-4$. We adopted the AdamW optimizer paired with a linear noise scheduler to optimize the training process.

4.2 Evaluation on Facial and Light Consistency

Single-view face relighting methods can be categorized into conventional and diffusion-based models. For each category, we select the recent SOTA relighting methods [15, 17, 25, 44] to ensure a comprehensive evaluation.

4.2.1 Quantitative Evaluations. We evaluate our relighting method on the Multi-PIE dataset [12] using two setups. In the first setup, the target lighting is sourced from the same individual, using 500 randomly selected input-target/groundtruth pairs from 250 subjects. In the second setup, the target lighting is taken from a different individual, using 250 randomly selected input-target-groundtruth triplets, where the target image comes from a different person.

Table 2: Evaluations on the same subject in Multi-PIE. The best performance is in bold and the second best is underlined.

Methods	LPIPS ↓	MSE ↓	DSSIM ↓	CSIM ↑	ITA ↓
Conventional methods					
DPR [44]	<u>0.1903</u>	0.0212	0.1405	0.7290	8.4782
Hou et al. (21) [17]	0.1998	0.0202	0.1445	0.7147	9.8010
Hou et al. (22) [15]	0.2003	0.0182	0.1324	<u>0.7577</u>	10.5299
Diffusion-based methods					
Difareli [25]	0.1942	<u>0.0125</u>	<u>0.1135</u>	0.7322	<u>6.7704</u>
Ours	0.1605	0.0078	0.0960	0.7793	6.1183

Table 3: Evaluations on the different subject in Multi-PIE. The best performance is in bold and the second best is underlined.

Methods	LPIPS ↓	MSE ↓	DSSIM ↓	CSIM ↑	ITA ↓
Conventional methods					
DPR [44]	<u>0.2018</u>	0.0308	0.1504	0.7177	10.0172
Hou et al. (21) [17]	0.2164	0.0251	0.1563	0.6873	11.8831
Hou et al. (22) [15]	0.2083	0.0220	0.1365	<u>0.7457</u>	10.8079
Diffusion-based methods					
Difareli [25]	0.2115	<u>0.0161</u>	<u>0.1243</u>	0.7150	<u>7.8863</u>
Ours	0.1725	0.0132	0.1050	0.7646	7.5919



Figure 7: Relighting results on the same and different subject in Multi-PIE. The skin tone for each result is in the top-right corner, and hard shadows are in the white box.

The results in Tab. 2 and Tab. 3 show that our IFS-Light method outperforms across all evaluated metrics. Compared to the recent SOTA diffusion-based method, Difareli [25], our model achieves notable improvements, with gains of 21.00%, 37.60%, 15.42%, 5.70%, and 9.63% in the same subject setup, and 14.52%, 18.02%, 15.53%, 6.94%, and 3.88% in the different subject setup. The significant reductions in LPIPS, MSE, and DSSIM underscore our model’s capability to produce high-quality relit images, while higher CSIM and lower ITA indicate enhanced consistency in facial and lighting conditions.

4.2.2 Qualitative Evaluations. We evaluate our relighting method on the Multi-PIE [12], FFHQ [18] and MST-E [30] datasets. Our evaluation set includes diverse subjects across pose, gender, skin tone, and accessories, with randomly chosen input-target pairs.

Relighting on Multi-PIE Dataset. In Fig. 7, we present relighting results on the same and different subject. For facial consistency, prior methods [15, 17, 25, 44] exhibit greater skin tone variations under lighting changes. Additionally, Difareli introduces noticeable details changes when lighting conditions vary significantly between input and reference images. In contrast, our method closely matches the ground truth with consistent skin tone (top-right corner) and fine details. In terms of lighting consistency, previous methods [15, 17, 25, 44] struggle to remove hard shadows when input images have pronounced shadows and reference images feature minimal ones (white boxes). IFS-Light effectively eliminates these shadows, owing to its spatial hard shadow coverage in the guidance.

Relighting on FFHQ Dataset. In Fig. 9, both DPR and Hou et al. (2021) frequently fail to accurately capture light intensity, leading to overexposed or underexposed relit images. Hou et al. (2022) often results in minimal lighting changes and may render skin tones with a reddish or blackish tint. Difareli, on the other hand, frequently lightens skin tones and modifies details (white box), such as hair, mouth, or glasses. Overall, our method demonstrates superior performance over these alternatives.

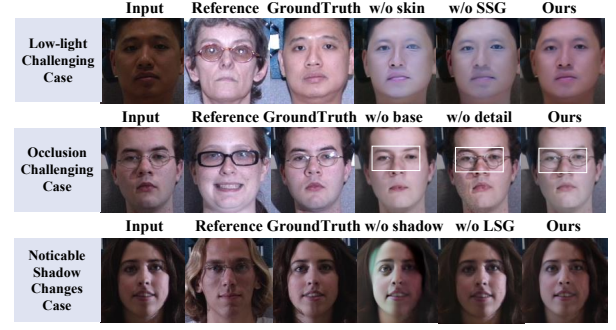


Figure 8: Ablation results of the challenging cases on Multi-PIE. The white box highlights specific details.

Relighting on MST-E Dataset. In Fig. 2 and Fig. 9, we present a compelling qualitative comparison of our method against recent methods. Our method consistently produces realistic results, effectively resolving lighting and facial inconsistencies, while competing methods often distort facial features or introduce shading artifacts in these in-the-wild images. For example, lighting adjustments in DPR and Hou et al. (2021) significantly alter the entire portrait, including the background. Similarly, Hou et al. (2022) and Difareli frequently generate inaccurate skin tones and noticeably change facial details, such as hair or cheeks. This contrast underscores the superior performance of our method, ensuring realistic lighting and facial portrayal across all subjects.

Table 4: Ablation studies with consistency conditions. The best performance is in bold and the second best is underlined.

Methods	LPIPS ↓	MSE ↓	DSSIM ↓	CSIM ↑	ITA ↓
Facial consistency conditions					
w/o skintone	0.1917	0.0137	0.1102	0.7313	7.8304
w/o SSG	0.1880	0.0142	0.1106	0.7232	<u>7.7876</u>
w/o detail	0.2166	0.0144	0.1202	0.7221	7.8325
w/o base	0.1887	<u>0.0134</u>	0.1097	0.7250	8.0123
Lighting consistency conditions					
w/o shadow	<u>0.1858</u>	<u>0.0134</u>	<u>0.1083</u>	0.7557	7.9533
w/o LSG	0.2078	0.0493	0.1628	0.8113	10.4977
Ours	0.1725	0.0132	0.1050	<u>0.7646</u>	7.5919

4.3 Ablation Studies

We evaluate our pipeline under four facial consistency conditions: w/o skin tone, w/o base features, w/o SSG, and w/o detail mask, along with two lighting consistency conditions: w/o shadow and w/o LSG. Results are shown in Tab. 4 and Fig. 8. Our full model achieves the best performance in four quality metrics, confirming that each component contributes to enhanced image quality.

Facial Consistency Conditions. In Tab. 4, IFS-Light shows superior performance across all metrics, notably achieving a 3.92% improvement in CSIM compared to w/o detail masks and a 4.16% reduction in ITA compared to w/o base features, underscoring the effectiveness of the proposed components in maintaining facial consistency. Fig. 8 shows that IFS-Light better preserves skin tones and details compared to other models in the cases of low light and occlusion.



Figure 9: Relighting results on FFHQ and MST-E. Skin tone is indicated in the top-right corner, while the white box highlights specific details. Note that the reference images of MST-E experiments are from Flickr [8].

Lighting Consistency Conditions. IFS-Light outperforms in lighting consistency in Tab. 4, with a notable 26.85% reduction in ITA compared to w/o LSG. However, since variations w/o LSG alter the original image less, these retain higher CSIM scores, as evidenced in the case of noticeable shadow changes in Fig. 8. The results confirm that lighting constraints are effective for enhancing consistency.

5 Conclusion

In this paper, we address two key challenges in single-view face relighting by introducing IFS-Light, an interactive framework that leverages spatial and non-spatial condition mechanisms for facial and lighting control. To ensure facial consistency, we propose a

skin-tone-scaled non-spatial condition combined with spatial skin-shape guidance to localize skin tone adjustments, further enhanced by a detail mask that refines editable facial regions while preserving background integrity. To maintain lighting consistency, we introduce a ray-tracing-based formulation to decompose reference lighting into non-spatial diffuse and non-diffuse components, which are integrated with shape information to generate spatial light-shadow guidance for precise lighting localization. Additionally, we propose an interactive light-ball editing way to define relighting results, enabling users to adjust light position for intuitive control over both intensity and direction, which offers a new interactive roadmap for the community. In future work, we will explore more user-friendly lighting control methods for single-view face relighting.

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References

- [1] Jonathan T Barron and Jitendra Malik. 2014. Shape, illumination, and reflectance from shading. *IEEE transactions on pattern analysis and machine intelligence* 37 (2014), 1670–1687.
- [2] Junuk Cha, Mengwei Ren, Krishna Kumar Singh, He Zhang, Yannick Hold-Geoffroy, Seunghyun Yoon, HyunJoon Jung, Jae Shin Yoon, and Seungryul Baek. 2024. Text2Relight: Creative Portrait Relighting with Text Guidance. *arXiv preprint arXiv:2412.13734* (2024).
- [3] Alain Chardon, Isabelle Cretois, and Colette Hourseau. 1991. Skin colour typology and suntanning pathways. *International journal of cosmetic science* 13 (1991), 191–208.
- [4] Sumit Chaturvedi, Mengwei Ren, Yannick Hold-Geoffroy, Jingyuan Liu, Julie Dorsey, and Zhixin Shu. 2025. SynthLight: Portrait Relighting with Diffusion Model by Learning to Re-render Synthetic Faces. *arXiv preprint arXiv:2501.09756* (2025).
- [5] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. 2019. Arcface: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4690–4699.
- [6] Haiwen Feng, Timo Bolkart, Joachim Tesch, Michael J Black, and Victoria Abreva. 2022. Towards racially unbiased skin tone estimation via scene disambiguation. In *European Conference on Computer Vision*. Springer, 72–90.
- [7] Yao Feng, Haiwen Feng, Michael J Black, and Timo Bolkart. 2021. Learning an animatable detailed 3D face model from in-the-wild images. *ACM Transactions on Graphics (ToG)* 40 (2021), 1–13.
- [8] Flickr. 2004. <https://www.flickr.com>
- [9] David Futschik, Kelvin Ritland, James Vecore, Sean Fanello, Sergio Orts-Escolano, Brian Curless, Daniel Šykora, and Rohit Pandey. 2023. Controllable light diffusion for portraits. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8412–8421.
- [10] Jian Gao, Chun Gu, Youtian Lin, Hao Zhu, Xun Cao, Li Zhang, and Yao Yao. 2023. Relightable 3d gaussian: Real-time point cloud relighting with brdf decomposition and ray tracing. *arXiv preprint arXiv:2311.16043* (2023).
- [11] Theo Gevers and Arnold WM Smeulders. 1999. Color-based object recognition. *Pattern recognition* 32 (1999), 453–464.
- [12] Ralph Gross, Iain Matthews, Jeffrey Cohn, Takeo Kanade, and Simon Baker. 2010. Multi-pie. *Image and vision computing* 28 (2010), 807–813.
- [13] Mingming He, Pascal Clausen, Ahmet Levent Taşel, Li Ma, Oliver Pilarski, Wenqi Xian, Laszlo Rikker, Xueming Yu, Ryan Burgert, Ning Yu, et al. 2024. DiffRelight: Diffusion-Based Facial Performance Relighting. In *SIGGRAPH Asia 2024 Conference Papers*. 1–12.
- [14] Yingqing He, Yazhou Xing, Tianjia Zhang, and Qifeng Chen. 2021. Unsupervised portrait shadow removal via generative priors. In *Proceedings of the 29th ACM International Conference on Multimedia*. 236–244.
- [15] Andrew Hou, Michel Sarkis, Ning Bi, Yiyong Tong, and Xiaoming Liu. 2022. Face relighting with geometrically consistent shadows. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4217–4226.
- [16] Andrew Hou, Zhixin Shu, Xuaner Zhang, He Zhang, Yannick Hold-Geoffroy, Jae Shin Yoon, and Xiaoming Liu. 2024. COMPOSE: Comprehensive Portrait Shadow Editing. In *European Conference on Computer Vision*. Springer, 356–373.
- [17] Andrew Hou, Ze Zhang, Michel Sarkis, Ning Bi, Yiyong Tong, and Xiaoming Liu. 2021. Towards high fidelity face relighting with realistic shadows. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 14719–14728.
- [18] Tero Karras, Samuli Laine, and Timo Aila. 2019. A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4401–4410.
- [19] Ha A Le and Ioannis A Kakadiaris. 2019. Illumination-invariant face recognition with deep relit face images. In *2019 IEEE Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 2146–2155.
- [20] Chen Li, Kun Zhou, and Stephen Lin. 2014. Intrinsic face image decomposition with human face priors. In *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V* 13. Springer, 218–233.
- [21] Tianye Li, Timo Bolkart, Michael J Black, Hao Li, and Javier Romero. 2017. Learning a model of facial shape and expression from 4D scans. *ACM Trans. Graph.* 36 (2017), 194–1.
- [22] Yaojie Liu, Andrew Z Hou, Xinyu Huang, Liu Ren, and Xiaoming Liu. 2022. Blind Removal of Facial Foreign Shadows.. In *BMVC*. 88.
- [23] Yiqun Mei, He Zhang, Xuaner Zhang, Jianming Zhang, Zhixin Shu, Yilin Wang, Zijun Wei, Shi Yan, HyunJoon Jung, and Vishal M Patel. 2023. LightPainter: interactive portrait relighting with freehand scribble. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 195–205.
- [24] Thomas Nestmeyer, Jean-François Lalonde, Iain Matthews, and Andreas Lehrmann. 2020. Learning physics-guided face relighting under directional light. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5124–5133.
- [25] Puntawat Ponglertnapakorn, Nontawat Tritrong, and Supasorn Suwajanakorn. 2023. DiFaReli: Diffusion face relighting. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 22646–22657.
- [26] Konpat Preechakul, Nattanat Chatthee, Suttisak Widadwongsa, and Supasorn Suwajanakorn. 2022. Diffusion autoencoders: Toward a meaningful and decodable representation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10619–10629.
- [27] Zia-ur Rahman, Daniel J Jobson, and Glenn A Woodell. 1996. Multi-scale retinex for color image enhancement. In *Proceedings of 3rd IEEE international conference on image processing*, Vol. 3. IEEE, 1003–1006.
- [28] Ravi Ramamoorthi and Pat Hanrahan. 2001. A signal-processing framework for inverse rendering. In *Proceedings of the 28th annual conference on Computer graphics and interactive techniques*. 117–128.
- [29] Mengwei Ren, Wei Xiong, Jae Shin Yoon, Zhixin Shu, Jianming Zhang, HyunJoon Jung, Guido Gerig, and He Zhang. 2024. Relightful Harmonization: Lighting-aware Portrait Background Replacement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 6452–6462.
- [30] Candice Schumann, Femi Olanubi, Auriel Wright, Ellis Monk, Courtney Hledreth, and Susanna Ricco. 2024. Consensus and subjectivity of skin tone annotation for ml fairness. *Advances in Neural Information Processing Systems* 36 (2024).
- [31] Soumyadip Sengupta, Angjoo Kanazawa, Carlos D Castillo, and David W Jacobs. 2018. Sfnet: Learning shape, reflectance and illuminance of faces in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 6296–6305.
- [32] Davoud Shahlaei and Volker Blanz. 2015. Realistic inverse lighting from a single 2d image of a face, taken under unknown and complex lighting. In *2015 11th IEEE international conference and workshops on automatic face and gesture recognition (FG)*, Vol. 1. IEEE, 1–8.
- [33] LuminaBrush Team. 2024. LuminaBrush: Illumination Drawing Tools for Text-to-Image Diffusion Models.
- [34] Unsplash. 2013. <https://unsplash.com>
- [35] Ao Wang, Hui Chen, Zijia Lin, Jungong Han, and Guiguang Ding. 2024. Replit: Revisiting mobile cnn from vit perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 15909–15920.
- [36] Xiaoyan Xing, Konrad Groh, Sezer Karaoglu, Theo Gevers, and Anand Bhatad. 2024. LumiNet: Latent Intrinsic Meets Diffusion Models for Indoor Scene Relighting. *arXiv preprint arXiv:2412.00177* (2024).
- [37] Xiaoyan Xing, Vincent Tao Hu, Jan Hendrik Metzen, Konrad Groh, Sezer Karaoglu, and Theo Gevers. 2024. Retinex-Diffusion: On Controlling Illumination Conditions in Diffusion Models via Retinex Theory. *arXiv preprint arXiv:2407.20785* (2024).
- [38] Jae Shin Yoon, Zhixin Shu, Mengwei Ren, Cecilia Zhang, Yannick Hold-Geoffroy, Krishna Kumar Singh, and He Zhang. 2024. Generative Portrait Shadow Removal. *ACM Transactions on Graphics (TOG)* 43 (2024), 1–13.
- [39] Changqian Yu, Jingbo Wang, Chao Peng, Changxin Gao, Gang Yu, and Nong Sang. 2018. Bisenet: Bilateral segmentation network for real-time semantic segmentation. In *Proceedings of the European conference on computer vision (ECCV)*. 325–341.
- [40] Jiangjian Yu, Ling Zhang, Qing Zhang, Qifei Zhang, Daiguo Zhou, Chao Liang, and Chunxia Xiao. 2024. Portrait Shadow Removal Using Context-Aware Illumination Restoration Network. *IEEE Transactions on Image Processing* (2024).
- [41] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. 2025. Scaling in-the-wild training for diffusion-based illumination harmonization and editing by imposing consistent light transport. In *The Thirteenth International Conference on Learning Representations*.
- [42] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. 2018. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 586–595.
- [43] Xuaner Zhang, Jonathan T Barron, Yun-Ta Tsai, Rohit Pandey, Xiuming Zhang, Ren Ng, and David E Jacobs. 2020. Portrait shadow manipulation. *ACM Transactions on Graphics (TOG)* 39 (2020), 78–1.
- [44] Hao Zhou, Sunil Hadap, Kalyan Sunkavalli, and David W Jacobs. 2019. Deep single-image portrait relighting. In *Proceedings of the IEEE/CVF international conference on computer vision*. 7194–7202.